

24MAT110 - ASSIGNMENT 1

1 Find the domain and range of the following functions

(a) $f(x) = 5 - \sqrt{x^2 - 2x - 3}$

Domain:

$$x^2 - 2x - 3 \geq 0$$

$$x^2 + x - 3x - 3 \geq 0$$

$$x(x+1) - 3(x+1) \geq 0$$

$$(x-3)(x+1) \geq 0$$

$$x \geq 3 \quad x \geq -1$$

~~so~~ so $(-\infty, -1] \cup [3, \infty)$ is the domain
of the function $f(x) = 5 - \sqrt{x^2 - 2x - 3}$

Range:

$$\sqrt{x^2 - 2x - 3} \geq 0$$

since square root should be non-negative

multiply -1 & add 5 both sides

$$-\sqrt{x^2 - 2x - 3} \geq 0$$

$$5 - \sqrt{x^2 - 2x - 3} \geq 5$$

$$f(x) \geq 5$$

$\therefore$ the range of the function is $(-\infty, 5)$

$$(b) f(x) = 2 + \frac{3x^2}{x^2 + 4}$$

$$x^2 + 4 \neq 0 \text{ in } \mathbb{R}$$

$$\therefore \text{Domain of function } f(x) = 2 + \frac{3x^2}{x^2 + 4} \\ \Rightarrow \mathbb{R}.$$



$$\text{Range of } f(x) = [2, 5)$$

2. Find the following limits without using L'Hospital's rule.

$$(a) \lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x - 1}$$

$$\Rightarrow \lim_{x \rightarrow 1} \frac{x^2 - x + 2x - 2}{x - 1}$$

$$\Rightarrow \lim_{x \rightarrow 1} \frac{(x-1)(x+2)}{x-1}$$

$$\Rightarrow \lim_{x \rightarrow 1} x + 2$$

$$\Rightarrow 1 + 2 = 3$$

$$(b) \lim_{x \rightarrow 1} \frac{x^{2/3} - 1}{1 - x^{1/2}}$$

$$x^{2/3} = (x^{1/6})^4 = y^4 \quad \text{let } x^{1/6} = y$$

$$x^{1/2} = (x^{1/6})^3 = y^3$$

$$\Rightarrow \lim_{y \rightarrow 1} \frac{y^4 - 1}{1 - y^3}$$

$$\Rightarrow \lim_{y \rightarrow 1} \frac{(y^2 - 1)(y^2 + 1)}{-(y^3 - 1)}$$

$$\Rightarrow \lim_{y \rightarrow 1} \frac{(y-1)(y+1)(y^2+1)}{-(y-1)(y^2+y+1)}$$

$$a^2 - b^2 = (a-b)(a+b)$$

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

$$\Rightarrow \lim_{y \rightarrow 1} \frac{(y+1)(y^2+1)}{-(y^2+y+1)}$$

$$\Rightarrow \frac{(1+1)(1^2+1)}{-(1^2+1+1)} = \frac{(2)(2)}{-(3)}$$

$$\Rightarrow -\frac{4}{3}$$

$$(C) \lim_{x \rightarrow 0} \frac{|2x+2| - |x| - 2}{x}$$

$$2x+2=0 \text{ when } x=-1$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{2x+2 - |x| - 2}{x} \text{ for values of } x \text{ in small neighborhood around zero}$$

$$2x+2 \text{ is +ve } -1 < x < \infty$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{2x - |x|}{x} \therefore \text{for } x \text{ near zero } |2x+2| = 2x+2$$

Right hand limit

$$\lim_{x \rightarrow 0^+} \frac{2x - x}{x} = \lim_{x \rightarrow 0^+} \frac{x}{x} = \lim_{x \rightarrow 0^+} 1 = 1$$

left hand limit

$$\lim_{x \rightarrow 0^-} \frac{2x - (-x)}{x} = \lim_{x \rightarrow 0^-} \frac{2x+x}{x} = \lim_{x \rightarrow 0^-} \frac{3x}{x} = \lim_{x \rightarrow 0^-} 3 = 3$$

$$RHL \neq LHL$$

$\therefore$ The limit does not exist

$$(d) \quad \lim_{x \rightarrow 0} \frac{x + x \cos x}{\sin x \cos x}$$

$$\Rightarrow \lim_{x \rightarrow 0} \left(\frac{x}{\sin x \cos x} + \frac{x \cos x}{\sin x \cos x} \right)$$

$$\Rightarrow \lim_{x \rightarrow 0} \left(\frac{x}{\sin x} \cdot \frac{1}{\cos x} + \frac{x}{\sin x} \right)$$

$$\boxed{\lim_{x \rightarrow 0} \frac{x}{\sin x} = 1}$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{1}{\cos x} + 1$$

$$\Rightarrow \frac{1}{\cos 0} + 1 = 1 + 1 = 2$$

$$(e) \quad \lim_{x \rightarrow 1} \frac{x^{50} - 1}{x - 1}$$

$$\Rightarrow \lim_{x \rightarrow 1} \frac{x^{49} \cdot x - x^{49} + x^{49} - 1}{x - 1}$$

$$\Rightarrow \lim_{x \rightarrow 1} \frac{x^{49} (x - 1) + x^{49} - 1}{x - 1}$$

$$\Rightarrow \lim_{x \rightarrow 1} \frac{x^{49} \cancel{(x-1)}}{\cancel{x-1}} + \lim_{x \rightarrow 1} \frac{x^{49} - 1}{x - 1}$$

$$\Rightarrow \lim_{x \rightarrow 1} x^{49} + \lim_{x \rightarrow 1} \left(\frac{x^{48} (x - 1) + x^{48} - 1}{x - 1} \right)$$

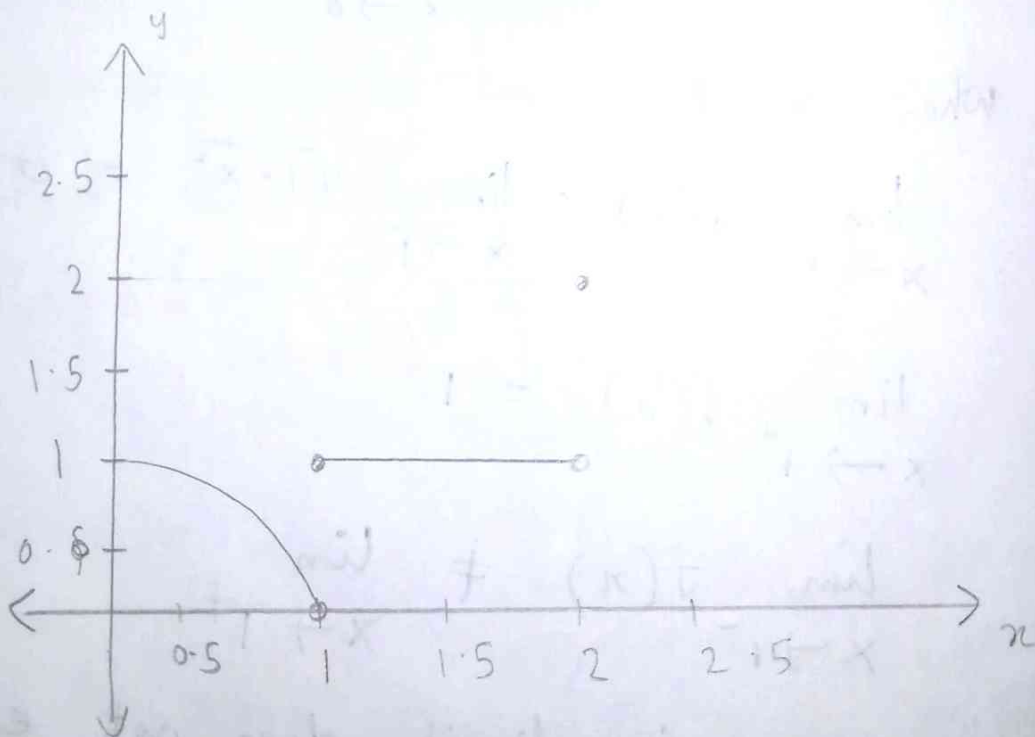
$$\Rightarrow \lim_{x \rightarrow 1} \lim_{x \rightarrow 1} x^{49} + \lim_{x \rightarrow 1} x^{48} + \dots + \lim_{x \rightarrow 1} x^2 + \lim_{x \rightarrow 1} x + \lim_{x \rightarrow 1} \left(\frac{x(x-1) + (x-1)}{x-1} \right)$$

$$\Rightarrow \lim_{x \rightarrow 1} (x^{49} + x^{48} + \dots + x^2 + x + 1)$$

$$\Rightarrow 49 + 1 = 50$$

3. Given a function $f(x) = \begin{cases} \sqrt{1-x^2} & 0 \leq x < 1 \\ 1 & 1 \leq x < 2 \\ 2 & x = 2 \end{cases}$

(a) Draw the graph of the function



b) Find the domain & Range of the function

$$\text{Domain : } [0, 2]$$

$$\text{Range : } [0, 2]$$

c) At what point does the limit exist? At what points does it not exist? Explain.

For $x=0$

$$\lim_{x \rightarrow 0^-} f(x) = \text{does not exist} \quad (f \text{ is not defined for } x < 0)$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \sqrt{1-x^2} = 1$$

$$\lim_{x \rightarrow 0} f(x) = 1 \quad \lim_{x \rightarrow 0} f(x) \text{ does not exist}$$

When $x=1$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} \sqrt{1-x^2} = 0$$

$$\lim_{x \rightarrow 1^+} f(x) = 1$$

$$\lim_{x \rightarrow 1^-} f(x) \neq \lim_{x \rightarrow 1^+} f(x)$$

$\therefore$ limit does not exist
for $x=1$

$$\lim_{x \rightarrow 1} f(x) \text{ does not exist}$$

At $x = 2$

$$\lim_{x \rightarrow 2^-} f(x) = 1$$

$\lim_{x \rightarrow 2^+} f(x)$ does not exist

(f is not defined for $x > 2$)

$\lim_{x \rightarrow 2} f(x)$ does not exist

$\therefore$ limit does not exist at the points

$$x = 1, 2$$

limit exists in the interval $(0, 1) \cup (1, 2)$

$$\text{for the } f(x) = \begin{cases} \sqrt{1-x^2} & 0 \leq x < 1 \\ \frac{1}{2} & 1 \leq x < 2 \\ x = 2 \end{cases}$$

4) Check the continuity of the function

$$f(x) = \begin{cases} -3x - 6 & x \leq 0 \\ 2x^2 - 6 & 0 < x \leq 2 \\ 3x - 5 & x > 2 \end{cases}$$

$$\text{At } x = 0 \Rightarrow \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} -3x - 6 = -6$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} 2x^2 - 6 = -6$$

$$f(0) = -3(0) - 6 = -6$$

$$\lim_{x \rightarrow 0^-} f(x) = f(0) = \lim_{x \rightarrow 0^+} f(x)$$

$$\therefore f(x) \text{ is continuous at } x = 0$$

At $x = 2$

$$\text{LHL } \lim_{x \rightarrow 2^-} f(x) = 2x^2 - 6 = 8 - 6 = 2$$

$$\text{RHL } \lim_{x \rightarrow 2^+} f(x) = 3x - 5 = 6 - 5 = 1$$

$$\text{LHL} \neq \text{RHL} \quad \lim_{x \rightarrow 2^-} f(x) \neq \lim_{x \rightarrow 2^+} f(x)$$

$\therefore \lim_{x \rightarrow 2} f(x)$ does not exist

$\therefore f(x)$ is not continuous at $x = 2$

5) Show that the function $f(x) = (x-a)^2(x-b)^2 + x$ takes on the value $\frac{a+b}{2}$ for some values x

$$f(x) = (x-a)^2(x-b)^2 + x$$

Using Intermediate Value Theorem

$$\begin{aligned} \lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} (x-a)^2(x-b)^2 + x \\ &= (a-a)^2(a-b)^2 + a \\ &= 0(a-b)^2 + a = a \end{aligned}$$

$$\begin{aligned}
 \lim_{x \rightarrow b} f(x) &= \lim_{x \rightarrow b} (x-a)^2(x-b)^2 + x \\
 &= (b-a)^2(b-b)^2 + b \\
 &= (b-a)^2 \cdot 0 + b = b
 \end{aligned}$$

$$\therefore f(x) \in [f(a), f(b)] \text{ for } x \in [a, b]$$

where $f(a) = a$, $f(b) = b$

$\Rightarrow$ there exists a ^{value of} x for which $f(x)$ would be equal to $\frac{a+b}{2}$

6) If the product function $h(x) = (x) \cdot g(x)$ is continuous at $x=0$, must (x) & $g(x)$ be continuous at $x=0$?

No (x) & $g(x)$ need not be continuous at $x=0$ if $h(x) = (x) \cdot g(x)$ is continuous at $x=0$

if $g(x)$ is a greatest integer function
 $h(x) = x \cdot [x]$

At $x = 0$ $\lim_{x \rightarrow 0^-} h(x) = \lim_{x \rightarrow 0^+} h(x) = 0$

$\therefore \lim_{h \rightarrow 0} h(x) = 0$

$h(x)$ is continuous even when $g(x)$ is not since it is greatest integer function.

1. Check the differentiability of the function $f(x) = \begin{cases} x^{2/3} & x \geq 0 \\ x^{1/3} & x < 0 \end{cases}$ at the origin

At $x = 0$ $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} x^{1/3} = 0$

$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} x^{2/3} = 0$

$\lim_{x \rightarrow 0} f(x) = 0$

LHL = RHL

$\therefore f(x)$ continuous at $x = 0$

at $x = 0$ $f'(x) = \frac{2}{3} x^{2/3-1} = \frac{2}{3} x^{-1/3}$

$= \frac{2}{3 x^{1/3}}$

$f'(0) = \frac{2}{3(0)^{1/3}} = \text{not defined}$

8. Find the two points where the curve $x^2 + xy + y^2 = 7$ crosses the x -axis, and show that the tangents to the curve at these points are parallel.

For the curve $x^2 + xy + y^2 = 7$ to cross the x -axis, $y = 0$

$$\therefore x^2 + x(0) + (0)^2 = 7$$

$$x^2 = 7$$

$$x = \pm\sqrt{7}$$

$\therefore$ The curve crosses the x -axis at

$$(-\sqrt{7}, 0) \text{ \& } (\sqrt{7}, 0)$$

$$x^2 + xy + y^2 = 7$$

$$\frac{d}{dx}(x^2 + xy + y^2) = \frac{d}{dx}(7)$$

$$2x + x \cdot \frac{dy}{dx} + y + 2y \frac{dy}{dx} = 0$$

$$2x + y + x \cdot \frac{dy}{dx} + 2y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx}(x + 2y) = -(2x + y)$$

$$\frac{dy}{dx} = \frac{-(2x + y)}{x + 2y}$$

tangent at $x = \sqrt{7}$

$$\left. \frac{dy}{dx} \right|_{x=\sqrt{7}} = \frac{-(2\sqrt{7} + 0)}{\sqrt{7} + 2(0)}$$

$$= -\frac{2\sqrt{7}}{\sqrt{7}} = -2$$

tangent at $x = -\sqrt{7}$

$$\left. \frac{dy}{dx} \right|_{x=-\sqrt{7}} = \frac{-(2(-\sqrt{7}) + 0)}{-\sqrt{7} + 2(0)} = -2$$

$\therefore$ slope of both tangents are equal

$\therefore$ tangents to the curve at these points are parallel.
